

Baseline Configuration and Controls in Metaheuristic Comparisons for Wind Farm Layout Optimization, with a PSO–VNS Reference Method

Prince Solanki, Pulkit Dwivedi, Vanita Garg, and Vinay Shukla

Abstract—Wind farm layouts fix a plant’s lifetime energy yield, yet comparisons of layout optimizers are hard to trust. We compare eight methods for continuous wind farm layout optimization under a controlled protocol (68 benchmark cases, 6,030 evaluations, feasibility-aware ranking, Holm-corrected and equivalence tests). First, baseline configuration changes the conclusions: the particle swarm optimization (PSO) setting $w = 0.7$, $c_1 = c_2 = 2$ violates the order-2 stability condition, so the swarm does not contract and most evaluations are infeasible; with it, PSO ranks behind salp-swarm hybrids; with the constriction setting it ranks first. Second, as the first phase before variable neighborhood search (VNS), a salp swarm (SSA) adds nothing beyond random sampling in the farm, a Laplacian SSA (LX-SSA) is worse (the Laplace step is harmful), and PSO gives a clear gain; at this budget, VNS acts as a local-search descent. Third, PSO-VNS (PSO followed by VNS) and a well-configured PSO are practically equivalent on average (within ± 0.05 percentage points of wake loss; posterior probability above 0.99); PSO-VNS adds feasibility reliability (Horns Rev 1 block: 30/30 vs 19/30 feasible runs) and gains in larger Data Set II layouts. These conclusions hold across qualification thresholds, farm clusters and one Holm family. PSO-VNS is a consistent reference method, not a superior one: VNS alone ranks first from feasible starts, and the best published IEA Wind Task 37 layouts remain better.

Index Terms—Benchmarking, equivalence testing, particle swarm optimization, variable neighborhood search, wake effect, wind farm layout optimization.

NOMENCLATURE

B	Budget (number of evaluations).
C_T, K	Thrust coefficient; Jensen wake-spreading constant.
D, R	Rotor diameter and rotor radius ($D = 2R$).
E_{ideal}	Wake-free benchmark objective of one turbine.
F_p	Penalized wake loss (minimized).

Manuscript received: XXXX; revised: XXXX; accepted: XXXX.

[TBD: funding statement, e.g., “This work did not receive any specific grant from funding agencies.”]

P. Solanki [TBD: mark corresponding author] is with the Department of Mathematics, School of Sciences, IILM University, Greater Noida, Uttar Pradesh, India; he was previously with the Department of Mathematics, Indian Institute of Technology Roorkee, Roorkee, India (e-mail: prince@ma.iitr.ac.in; prince@iilm.edu).

P. Dwivedi is with the Department of Computer Science and Engineering, Bennett University, Greater Noida, Uttar Pradesh, India (e-mail: pdwivedi19900@gmail.com; pulkit.dwivedi@bennett.edu.in).

V. Garg is with Amity University, Noida, Uttar Pradesh, India (e-mail: vanitagarg16@gmail.com).

V. Shukla is with the School of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai 200240, China, and also with the Department of Mathematics, School of Computer Science Engineering and Technology, Bennett University, Greater Noida, India (e-mail: vinayshukla4321@gmail.com; vnsukla01@sjtu.edu.cn).

$\ell_{\min}$	Minimum spacing between turbines.
N, N_p	Number of turbines; population size.
N_θ, N_s	Numbers of wind-direction and wind-speed bins.
r	Farm radius.
w, c_1, c_2	Inertia weight and acceleration coefficients of PSO.
$\mathbf{X}$	Layout vector $(x_1, y_1, \dots, x_N, y_N)$.
θ	Wind direction (blowing towards, CCW from east).
ω	Fraction of the budget given to Phase 1.

I. INTRODUCTION

The positions of the turbines in a wind farm are fixed at the planning stage and determine, for its lifetime, how much of the available wind energy it converts. An upstream turbine leaves behind a wake of slower and more turbulent air that reduces the power of the turbines downstream, so the layout sets the wake loss and thereby the gross energy yield, and shapes the collector system and grid connection [1], [2]. As the wake loss is a share of the wake-free energy, a difference of a few tenths of a percentage point is a lifetime energy difference of that size. The wind farm layout optimization problem (WFLOP) places a given number of turbines in a site to minimize the wake loss subject to boundary and minimum-spacing constraints; we address only this wake-loss part of layout design, for farms of up to 36 turbines.

Early studies placed turbines on a grid with genetic algorithms (GAs) [3], [4]. The continuous model of Kusiak and Song [5], a common reference for metaheuristics, has been solved with ant colony optimization and particle filtering [6], [7] and biogeography-based optimization [8]; GA, particle swarm optimization (PSO), differential evolution (DE) and simulated annealing were compared in [9], and PSO has reduced wake losses in [10]. Later work used informed GAs [11], [12], deterministic refinement [13], variable neighborhood search (VNS) [14], pseudo-gradients [15], Gaussian wake models [16] and surrogates [17]–[19].

Guidelines for comparing metaheuristics are well established: controlled rather than competitive experiments [20], equal budgets, reported settings, component analysis and paired nonparametric tests with multiplicity correction [21]–[24], and equivalence or Bayesian analyses of practically relevant differences [25], [26]. Several metaphor-based algorithms, the salp swarm algorithm (SSA) [27] among them,

reuse known search operators [28]–[30]. In the WFLOP, the GECCO layout competition [31] and the IEA Wind Task 37 studies [32], [33] compared optimizers on a common objective, and reviews list many metaheuristics tested under differing budgets and settings [34]; these studies compare methods as submitted or as published, not how the configuration of a baseline or the choice of control changes which method appears best. Four issues matter here. First, the PSO setting $w = 0.7$, $c_1 = c_2 = 2$ (inertia weight and acceleration coefficients), which keeps the acceleration coefficients of the original PSO [35], lies in the order-1 (mean) stability region but outside the smaller order-2 (variance) region [36]–[38]. Second, budgets counted in iterations, or without the gradient evaluations of gradient-based solvers, are not equal. Third, infeasible layouts returned by penalty-based metaheuristics must be ranked consistently. Fourth, mean values without paired tests and multiplicity correction [39] may mislead, and a nonsignificant difference is not evidence of equivalence [26]. For two-phase hybrids, a further question is whether the first phase contributes more than a random start would.

Our thesis is that the conclusions of such comparisons depend strongly on how the baselines are configured and how the comparison is controlled. Under a controlled protocol, a simple two-phase design, PSO followed by the basic VNS [40], [41] (PSO-VNS), and a well-configured PSO rank first and second of eight methods and are practically equivalent on average, while salp-swarm phases add nothing beyond random sampling and the Laplace step of the Laplacian SSA (LX-SSA) [42], proposed earlier by two of the authors, makes the hybrid worse. PSO-VNS is a sequential hybrid in the memetic sense, not a new PSO variant. Our contributions are:

- 1) The application of established practice [22]–[24] to the continuous WFLOP (equal budgets counting gradient evaluations, seed-paired runs, case-mean and run-level tests with Holm correction, model re-evaluation, public code and data), with three elements not reported for this problem to our knowledge: feasibility-aware ranking, random-sampling controls of equal cost, and an equivalence analysis (two one-sided tests and a Bayesian signed-rank test [25], [26]) with a margin of ± 0.05 percentage points (pp) of wake loss, fixed after the primary analysis and reported with the minimal equivalence margin. The main conclusions hold across qualification thresholds, farm clusters and one Holm family over all main-text tests. We apply it to eight methods on 68 benchmark cases, a 16-turbine block of the Horns Rev 1 farm and IEA Wind Task 37 (IEA37) Case Study 1 (Sections V, VIII and IX).
- 2) Evidence, with its mechanism, that the PSO setting changes the order of PSO and the salp-swarm methods. The old setting violates the order-2 condition (Proposition 1); its swarm stays 8.9 times wider than with the constriction setting [43], and late in the run only 2.6% of its evaluations are feasible (constriction: 48.4%). PSO is then feasible in 86.0% of the runs and ranks behind both salp-swarm hybrids of Table I; with constriction it ranks first there and second in the main comparison,

ahead of every salp-swarm method, at 6,030 evaluations (at 120,030, SSA-VNS ranks ahead of PSO on the six largest cases; Sections III-A, VI-A and VIII-A).

- 3) A component analysis of two-phase swarm-VNS hybrids against equal-cost random sampling in the farm disc (RSD-VNS) and its bounding square (RS-VNS), a weak control: a square sample is feasible with probability $\leq (\pi/4)^N$ (Proposition 2); 730 of its 2,040 runs lack a feasible sample (RSD-VNS: 494). Salp-swarm phases add nothing beyond RSD-VNS: SSA-VNS gains nothing (the case means, not the clusters, favor RSD-VNS: +0.024 pp) and LX-SSA-VNS is worse (+0.087 pp); the small SSA gain over RS-VNS reflects only that control's weakness. The Laplace step is harmful ($p = 5.4 \times 10^{-6}$); a PSO phase clearly beats both controls (−0.306 pp vs. RSD-VNS). At this budget, the VNS phase is one compass-search descent (the analysis values a local-search phase); it significantly improves both salp-swarm methods but PSO only slightly (Section VII).
- 4) PSO-VNS as a consistent reference method, not a superior one. It has the best average rank (1.82) and is feasible in every run, but it is practically equivalent to a well-configured PSO on average (90% interval of the mean case difference [−0.041, 0.004] pp; minimal equivalence margin 0.041 pp; Bayesian probability of practical equivalence above 0.99). It adds feasibility reliability (Horns Rev 1 block: 30/30 vs 19/30 feasible runs) and gains in the larger layouts ($N \geq 10$, post hoc) of Data Set II (10 of 10 cases lower, $p_{\text{Holm}} \leq 0.018$). It is not first from feasible starts (VNS), under a Gaussian re-evaluation (PSO) or against the best published IEA37 layouts (Sections VIII and IX). From these results we derive benchmarking recommendations (Section X).

II. PROBLEM MODELING

For a prescribed number N of identical turbines, the $n = 2N$ coordinates (x_i, y_i) are optimized to maximize the expected farm power, i.e., to minimize the wake loss, under the Jensen wake model [44], [45] and the power curve of [5]. The wind direction θ has the probability density p_θ , and for each direction the wind speed s follows a Weibull density $p_s(s; \zeta, \psi)$ with shape $\zeta(\theta)$ and scale $\psi(\theta)$. Here θ is the direction *towards* which the wind blows, counter-clockwise from east; the meteorological directions θ_{met} of Horns Rev 1 and IEA37 (wind *from*, clockwise from north) are converted by $\theta = 270^\circ - \theta_{\text{met}}$. Two constraints apply: every pair of turbines satisfies $\ell_{ij} = \|(x_i - x_j, y_i - y_j)\|_2 \geq \ell_{\min} = 4D = 8R$, as in [5], [8], and the farm is circular, $x_i^2 + y_i^2 \leq r^2$, with $r = 500$, 750 or 1000 m.

A. Wake Effect Model

In the Jensen model (Figs. S15–S17 of the supplementary material), the wake of turbine j is a half cone with half-angle $\alpha = \arctan K$ (wake-spreading constant K), and its virtual vertex lies R/K upstream of the rotor. With $u_{ij} = (x_i - x_j) \cos \theta + (y_i - y_j) \sin \theta + R/K$ and the lateral offset $v_{ij} = -(x_i - x_j) \sin \theta + (y_i - y_j) \cos \theta$, turbine i lies in the wake if

$\beta_{ij} = \cos^{-1}(u_{ij}/\sqrt{u_{ij}^2 + v_{ij}^2}) < \alpha$. Its downstream distance is $d_{ij} = |u_{ij} - R/K|$, and with the thrust coefficient C_T the velocity deficit is

$$\Delta_{ij} = 1 - \frac{s_{\text{down}}}{s_{\text{up}}} = \frac{1 - \sqrt{1 - C_T}}{(1 + K d_{ij}/R)^2}. \quad (1)$$

As in [5], this top-hat wake is tested at the hub center only, so the deficit jumps at the cone edge, and it also reaches a turbine less than R/K upstream of j near its axis (non-physical, kept for comparability). The deficits are combined by the root-sum-square rule,

$$\Delta_i(\theta) = \left(\sum_{j \neq i, \beta_{ij} < \alpha} \Delta_{ij}^2 \right)^{1/2}. \quad (2)$$

As C_T is constant (which overstates the deficits above rated speed), the local wind speed in a direction bin is the free-stream speed times $1 - \Delta_i(\theta)$, a Weibull variable with shape $\zeta(\theta)$ and scale $\psi_i(\theta) = \psi(\theta)[1 - \Delta_i(\theta)]$.

B. Power Model

The power curve of the Kusiak–Song benchmark [5] is $f(s) = 0$ for $s < s_{\text{cut-in}}$, $f(s) = \lambda_1 s + \lambda_0$ for $s_{\text{cut-in}} \leq s \leq s_{\text{rated}}$ and $f(s) = P_{\text{rated}}$ above, with $s_{\text{cut-in}} = 3.5$ m/s, $s_{\text{rated}} = 14$ m/s, $P_{\text{rated}} = 1500$ kW, $\lambda_1 = 140.86$ and $\lambda_0 = -500$. We use it as published: it is slightly negative just above the cut-in speed, jumps from 1472 to 1500 kW at the rated speed and has no cut-out speed. The expected power of turbine i ,

$$E(P_i) = \int_0^{360} p_\theta(\theta) \int_0^\infty f(s) p_s(s; \zeta(\theta), \psi_i(\theta)) ds d\theta, \quad (3)$$

is evaluated, as in [5], by a Riemann sum over N_θ direction bins with probabilities p_l and N_s speed bins between $s_{\text{cut-in}}$ and s_{rated} (Section S-VIII-A of the supplementary material). The direction sum keeps the bin width in degrees as a factor; with 24 bins of 15° and $\sum_l p_l = 1$, the benchmark objective $\sum_i E(P_i)$ is thus 15 times the expected farm power in kW, and the gross (wake-only) AEP of the benchmark is $\text{AEP}[\text{MWh}] = (\text{objective}/15) \times 8.76$. For Data Set I, the wake-free benchmark objective of one turbine, $E_{\text{ideal}} = 14,045.74$, corresponds to 936.38 kW (a capacity factor of 62%, far above real sites) or 8,202.7 MWh/yr.

C. Constraint Handling

After every position update, each coordinate is clipped to $[-r, r]$ (box repair; the square, not the circle). The constraints are handled by a penalty on the violations $g_i^b = x_i^2 + y_i^2 - r^2$ and $g_{ij}^s = \ell_{\min} - \ell_{ij}$; all optimizers except MS-SLSQP minimize the penalized wake loss

$$F_p(\mathbf{X}) = \left(N E_{\text{ideal}} - \sum_{i=1}^N E(P_i) \right) + \sum_{i: g_i^b > 0} (1 + \mu g_i^b)^2 + \sum_{i < j: g_{ij}^s > 0} (1 + \mu g_{ij}^s)^2, \quad \mu = 10^{10}, \quad (4)$$

where $\mathbf{X} = (x_1, y_1, \dots, x_N, y_N)$; since $N E_{\text{ideal}}$ is the wake-free benchmark objective, the first term is the wake loss reported in our tables. Any violation above about 10^{-7} (m or m²)

makes the penalty exceed the largest possible wake loss, so, up to smaller violations, comparing F_p values is a feasibility-first rule: a feasible layout beats an infeasible one, feasible layouts are compared by wake loss, and infeasible ones by total penalty. As g_i^b is in m² and g_{ij}^s in m, the penalty favours infeasible layouts inside the farm with turbines too close. MS-SLSQP instead minimizes the wake loss under explicit constraints. A final layout counts as feasible if $\ell_{ij} \geq \ell_{\min}$ and $\sqrt{x_i^2 + y_i^2} \leq r$ hold within 10^{-6} m (Section V).

III. PRELIMINARIES: PHASE-1 METHODS AND BASIC VNS

A population method keeps N_p candidate vectors $\mathbf{X}^i \in \mathbb{R}^n$ with components X_m^i and bounds lb_m, ub_m ; unless stated otherwise, one iteration costs N_p objective evaluations.

A. PSO and Its Stability Conditions

We use the canonical global-best PSO [35] in the inertia-weight form [46] (the standard PSO of [47] uses a ring topology instead). Particle i has a position $\mathbf{X}^i$, a velocity $\mathbf{V}^i$ and a personal best $\mathbf{P}^i$, the swarm keeps the global best $\mathbf{G}$, and a particle moves by

$$\mathbf{V}^i \leftarrow w \mathbf{V}^i + c_1 \rho_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2 \rho_2 \circ (\mathbf{G} - \mathbf{X}^i), \quad (5)$$

$$\mathbf{X}^i \leftarrow \mathbf{X}^i + \mathbf{V}^i, \quad (6)$$

where ρ_1, ρ_2 are vectors of independent $U(0, 1)$ numbers, drawn anew for every move, and $\circ$ is the componentwise product. The constriction PSO of Clerc and Kennedy [43] with their standard choice $\varphi_1 = \varphi_2 = 2.05$ (constriction factor $\kappa = 0.7298$) is identical to (5) with

$$w = 0.7298, \quad c_1 = c_2 = \kappa \varphi_1 = 1.49618. \quad (7)$$

PSO denotes (5), (6) with these coefficients, fixed a priori from the literature, and the *old setting* the same algorithm with $w = 0.7$, $c_1 = c_2 = 2$, i.e., the acceleration coefficients of the original PSO [35] combined with an inertia weight. Velocities start at zero and are not clamped; a coordinate clipped to the square (Section II-C) keeps its velocity (no reflection or damping); $\mathbf{P}^i$ and $\mathbf{G}$ are updated after each evaluation.

Proposition 1: Under stagnation (fixed $\mathbf{P}^i, \mathbf{G}$) and without active bounds, the expected position converges to $(c_1 \mathbf{P}^i + c_2 \mathbf{G}) / (c_1 + c_2)$ for all initial values if and only if $|w| < 1$ and $0 < c_1 + c_2 < 4(1 + w)$ (order-1 stability; $(c_1 + c_2)/2 = E(c_1 \rho_1 + c_2 \rho_2)$). For $c_1 = c_2$, the second moments converge as well (order-2 stability), with a limit variance $\propto (P_m^i - G_m)^2$ in coordinate m , if and only if, in addition [36],

$$c_1 + c_2 < \frac{24(1 - w^2)}{7 - 5w}. \quad (8)$$

Proofs, also for $c_1 \neq c_2$ (stricter bound): Section S-VI-A of the supplementary material; the region also holds under a weaker, non-stagnant assumption [38]. The constriction setting is order-2 stable ($c_1 + c_2 = 2.99 < 3.35$); the old setting is order-1 stable ($4 < 6.8$) but not order-2 stable ($4 > 3.50$); its second moments grow by the factor 1.12 per iteration (Fig. S13). Neither condition implies that $\mathbf{G}$ is good. With bounds, a turbine with a clipped coordinate lies outside the farm (Lemma S3). In our runs (Fig. S14), the old swarm

does not contract (final spread 8.9 times, speed 13 times the constriction values), clips 8.7% of the coordinates per iteration (constriction 1.6%) and evaluates few feasible layouts (2.6% vs. 48.4% in the second half); the 282 infeasible final layouts of its runs mostly violate only the spacing (246; Section II-C); the setting changes the ranking (Section VI-A). Likewise, the DE scale factor 0.5 (Section V) exceeds Zaharie's critical value $\sqrt{(1 - \text{CR}/2)/N_p} = 0.14$ (CR = 0.9) [48].

B. Salp Swarm Algorithm and Its Laplacian Variant

The SSA [27] keeps the best solution found so far as the food source $\mathbf{H}$ and splits the sorted population into leaders and followers. A leader $i \leq N_p/2$ updates each coordinate by

$$X_m^i = H_m \pm \xi_1((ub_m - lb_m)\xi_2 + lb_m), \quad (9)$$

with either sign with probability 1/2, $\xi_2 \sim U(0, 1)$ and $\xi_1 = 2 \exp[-(4t/T)^2]$ decreasing with the iteration t out of T ; a follower $i > N_p/2$ moves to the midpoint of its position and the new position of the preceding salp. The leaders thus perturb $\mathbf{H}$ uniformly with a shrinking radius, and the followers only average [30]. LX-SSA [42] gives each follower a second candidate $Y_m^i = X_m^i + \gamma(H_m - X_m^i)$ with the Laplace step [49]

$$\gamma = \phi - \chi \operatorname{sgn}(z - \frac{1}{2}) \ln(1 - 2|z - \frac{1}{2}|), \quad z \sim U(0, 1), \quad (10)$$

$\phi = 0$, $\chi = 1$, and keeps the better candidate. As published, it evaluates both candidates of each follower and then re-evaluates the kept ones with the new population, so an iteration costs $2N_p$ evaluations and LX-SSA runs half as many iterations as SSA; comparing the two tests LX-SSA as published, not the Laplace step alone. Pseudocode: Section S-I-B of the supplementary material.

C. Basic Variable Neighborhood Search

The basic VNS [40], [41], in the continuous form of [50], improves one incumbent $\mathbf{X}$ with $k_{\max} = 5$ nested ℓ_∞ neighborhoods $\mathcal{N}_k(\mathbf{X}) = \{\mathbf{X}' : \delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \leq \delta_k\}$, $\delta_k = 0.1kr$, of the whole layout. Shaking draws a point uniformly from $\mathcal{N}_k(\mathbf{X})$ (box-repaired), and a best-improvement local search improves it. As the top-hat wake makes F_p discontinuous, this is a compass search: it evaluates all $2n$ moves $\pm h \mathbf{e}_m$ ($\mathbf{e}_m$ the m th unit vector), moves to the best improving one, and halves h (from $0.05r$) when none improves, until $h < 10^{-3}r$.

IV. TWO-PHASE SWARM-VNS DESIGN AND PSO-VNS

A. Two-Phase Template

All hybrids here follow one sequential (relay) template. For a budget of B evaluations and a split $\omega \in (0, 1)$, Phase 1 starts from the seeded initial population, runs a population method for about ωB evaluations and passes its best layout to Phase 2, the basic VNS, which applies the local search to it and then iterates until B evaluations are used. Only Phase 1 differs: PSO in *PSO-VNS*; SSA or LX-SSA in *SSA-VNS* and *LX-SSA-VNS* (with ξ_1 compressed to the Phase-1 iterations, so, unlike PSO, not the start of a stand-alone run); and $\text{round}(\omega B)$ layouts drawn uniformly in the bounding square, including the initial

Algorithm 1 PSO-VNS for a prescribed turbine count N

Require: budget B , split ω , N_p , w , c_1 , c_2 , radius r , $k_{\max}$

- 1: Draw and evaluate N_p particles $\mathbf{X}^i$; $\mathbf{V}^i \leftarrow \mathbf{0}$; $\mathbf{P}^i \leftarrow \mathbf{X}^i$;
 $\mathbf{G} \leftarrow \arg \min_i F_p(\mathbf{P}^i)$ ▷ Phase 1
- 2: **for** $t = 1, \dots, T_1$ and $i = 1, \dots, N_p$ **do**
- 3: Move $\mathbf{X}^i$ by (5), (6); box repair; evaluate $F_p(\mathbf{X}^i)$;
 update $\mathbf{P}^i$, $\mathbf{G}$
- 4: **end for**
- 5: $\mathbf{X} \leftarrow \text{LOCALSEARCH}(\mathbf{G})$; $k \leftarrow 1$ ▷ Phase 2
- 6: **while** fewer than B evaluations have been used **do**
- 7: $\mathbf{X}' \leftarrow \text{LOCALSEARCH}(\text{uniform point of } \mathcal{N}_k(\mathbf{X}))$
- 8: **if** $F_p(\mathbf{X}') < F_p(\mathbf{X})$ **then** $\mathbf{X} \leftarrow \mathbf{X}'$, $k \leftarrow 1$ **else** $k \leftarrow$
 $k \bmod k_{\max} + 1$
- 9: **end while**
- 10: **return** best layout evaluated

population, in the control *RS-VNS*. VNS alone is the limiting case in which Phase 1 is the initial population. PSO and SSA run for $T_1 = \text{round}((\omega B - N_p)/N_p)$ Phase-1 iterations, LX-SSA for $\text{round}((\omega B - N_p)/(2N_p))$. A swarm phase is useful only if it gives VNS a better start than random sampling with about the same number of evaluations (at $\omega = 0.5$, 3,015 for RS-VNS and $N_p(T_1 + 1) = 3,030$ for the swarms); the component analysis (Section VII) tests this for each Phase-1 choice, with RS-VNS as a weak, no-search minimum bar.

Proposition 2: A layout drawn uniformly in the bounding square has all N turbines inside the farm with probability $(\pi/4)^N$; its feasibility probability is smaller (Proposition S5). For the largest cases, it is below 10^{-6} (Table S34): Phase 1 of RS-VNS almost never contains a feasible layout (none in 730 of its runs) and passes its least-violating sample to VNS.

B. PSO-VNS

PSO-VNS (Algorithm 1) is not a new PSO variant: Phase 1 is the plain PSO of Section III-A. Since the coefficients do not depend on the iteration, Phase 1 reproduces exactly, for a given seed, the first $N_p(T_1 + 1)$ evaluations of a PSO run, so comparing PSO-VNS with PSO isolates the use of the second part of the budget.

PSO-VNS uses $N_p = 30$, the coefficients (7) and the VNS settings above, and adds one parameter, the split ω . With the default $\omega = 0.5$ and $B = 6,030$, PSO runs for $T_1 = 100$ iterations (99.5 rounded up) and VNS receives the remaining 3,000 evaluations. None of these values was tuned; Section VII studies ω . At this budget, Phase 2 is essentially one compass-search descent from $\mathbf{G}$ (median 1 shaking step, 0.02% of the Phase-2 evaluations; the first descent gives 97.5% of the Phase-2 gain and is unfinished in 39 of 40 runs with $N \geq 12$; Section VII); its neighborhoods matter only for small N .

V. EVALUATION PROTOCOL

A. Cases and Methods

The 68 cases combine the two Kusiak–Song wind data sets [5], [8] (Table S1) with circular farms of radii $r = 500$, 750 and 1000 m and $N = 2, \dots, 10$, $2, \dots, 12$ and $2, \dots, 15$

turbines ($R = 38.5$ m, $\ell_{\min} = 4D = 308$ m, $C_T = 0.8$, $K = 0.075$).

The *main comparison* includes eight methods ($N_p = 30$ where applicable): PSO-VNS (Algorithm 1, $\omega = 0.5$); PSO; SSA-VNS; SSA; LX-SSA; DE/rand/1/bin [51] (scale factor 0.5, crossover rate 0.9); VNS; and MS-SLSQP, a multi-start sequential least-squares programming method (SciPy's SLSQP [52], forward-difference gradients). All methods except MS-SLSQP minimize F_p (4) over $[-r, r]^{2N}$; MS-SLSQP minimizes the wake loss under explicit constraints, restarting from the initial layouts and then from random ones, and returns its best feasible layout (settings: Section S-I). The *component analysis* (Section VII) adds LX-SSA-VNS and the no-search control RS-VNS (Section IV-A), whose 3,015 random layouts cost about as much as the 3,030 Phase-1 evaluations of the hybrids; a split study varies ω on twelve cases (Section VII).

B. Comparison Design and Statistics

Equal budgets and paired seeds. Unless stated otherwise, a run uses exactly $B = 6,030$ objective evaluations, counted by an objective wrapper, including the initial population and MS-SLSQP's gradient evaluations. Each method uses seeds 1–30 and draws its initial population first, so runs with equal seeds share the initial population and are paired (where logged, the best initial objectives agree, also with the earlier version).

Feasibility-aware ranking. Within a case, a method is ranked by the mean benchmark objective of its feasible runs only if at least 15 of its 30 runs (5 of 10 for 10 seeds) are feasible, and otherwise below all such methods, by its number of feasible runs. This limits, but does not remove, survivorship bias.

Statistics. All tests are two-sided ($\alpha = 0.05$), with Holm's correction [53] within each family of comparisons. 1) Case ranks: the Friedman test with the Iman–Davenport F test and post hoc tests of the average ranks against PSO-VNS [21], [22], [24]. 2) Case means (primary tests): since mean-rank post hoc tests depend on the pool [39], Wilcoxon signed-rank tests of PSO-VNS against each method on the per-case mean wake losses; a case in which only one of the two methods has at least 15 feasible runs counts as a maximal difference in its favor (2 for SSA, 2 for LX-SSA and 18 for DE cases), and one in which neither has is dropped (0 cases). Headline mean differences (cases in which both qualify) carry 95% bootstrap confidence intervals over cases. 3) Runs: per case, Wilcoxon signed-rank tests on the 30 seed-paired runs (infeasible runs below all feasible ones, ordered by minimum spacing), Holm-corrected over the comparisons of the case, with the matched-pairs rank-biserial correlation r_{rb} (zero differences dropped) and descriptive counts of significant wins, ties and losses (W/T/L). 4) Equivalence, as nonsignificance is not equivalence [26]: TOST on case-mean differences (90% bootstrap interval within ± 0.05 pp, 0.05% of the wake-free energy; fixed after the primary analysis, minimal margin reported) and a Bayesian signed-rank test with this region [25] (Table S13). Jensen and Gaussian wake losses of the same layouts differ by 0.38 pp on average (7.6 times the margin), but the PSO-VNS–PSO difference changes by only 0.031 pp (Table S23).

Robustness of inference. The 68 cases form six (data set, radius) clusters with nested N , not independent draws. Thresh-

olds of 1, 10, 15, 20, 25 and 30 feasible runs change no verdict; of the 68-case comparisons, cluster analyses (sign per cluster, leave-one-out) qualify only SSA-VNS vs. RSD-VNS and LX-SSA-VNS vs. RSD-VNS; one Holm family over all 28 case-mean tests of the paper makes only PSO-VNS vs. PSO, $N \geq 10$, both data sets and PSO-VNS ($\omega = 0.5$) vs. PSO-VNS ($\omega = 0.9$) nonsignificant (Tables S15–S20).

C. Tests Beyond the Benchmark and Reproducibility

Initialization and budget. Random initial layouts are usually infeasible; the feasible initialization draws seeded layouts inside the site and separates the turbines with L-BFGS-B on a packing penalty, without objective evaluations (Section S-I); all methods but RS-VNS (random sampling) use it at 6,030 evaluations. With random starts, all ten run at 6,030, 30,030 and 120,030 evaluations. Both studies use the six largest cases (largest N per farm and data set) and the Horns Rev 1 block, with 30 seeds (10 for Horns Rev 1 at 120,030).

Other sites and models. The Horns Rev 1 block is PyWake's 16-turbine example [54]: V80 turbines ($D = 80$ m), the measured 12-sector wind climate, the parallelogram through the corner turbines as boundary (Section VIII-B), $\ell_{\min} = 4D$ and the Jensen wake with $K = 0.04$. The 16- and 36-turbine scenarios of IEA37 Case Study 1 [32], [55] (circular sites of radius 1300 and 2000 m, $\ell_{\min} = 2D$, 16 directions at one speed, a simplified Bastankhah Gaussian wake) test optimizers rather than represent a site; all ten methods run on them at 6,030 and 30,030 evaluations (30 seeds).

Verification and code. Our objective matches the original implementation (largest difference 8.7×10^{-11} over 720 archived final objectives) and the official IEA37 calculator (relative difference $< 10^{-11}$). Our reused code of SSA, LX-SSA, PSO and DE (also in the hybrids) reproduces the result distributions of the earlier version (2 of 24 Mann–Whitney tests reject at $\alpha = 0.05$; 1.2 expected by chance). Instrumented copies for the diagnostics reproduce the stored runs and match the original code bit for bit (62 reruns). The runs used Linux containers (Intel Xeon, 2.1 GHz, 4 virtual cores; Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1); code, per-run results and scripts are public (Data Availability).

VI. RESULTS ON THE BENCHMARK

This section reports the 68 benchmark cases (22,440 runs, 6,030 evaluations each); per-case means and standard deviations are given in Tables S3–S8 of the supplementary material.

A. Effect of the Baseline Setting

The old setting violates the order-2 (variance) stability condition (8). Run on the same cases with the same budget and seeds as the constriction setting (7), it returns a feasible layout in only 86.0% of the runs (fewer than 15 feasible runs in 8 cases), is significantly worse in 56 of the 68 cases and never better (Wilcoxon, 30 seed-paired runs), and has a mean wake loss 0.61 percentage points higher over the 60 cases in which both settings have at least 15 feasible runs. Instrumented reruns show the mechanism (Table S35): with the old setting

TABLE I

EFFECT OF THE PSO SETTING ON THE METHOD POOL OF THE OLD-SETTING COMPARISON (THE POOL OF TABLE II WITH LX-SSA-VNS INSTEAD OF PSO-VNS; 68 CASES, 6,030 EVALUATIONS): AVERAGE RANK (1 = BEST) WITH THE OLD ($w = 0.7$, $c_1 = c_2 = 2$) AND THE CONSTRICTION SETTING, AND RUN-LEVEL W/T/L OF THE SALP-SWARM HYBRIDS AGAINST PSO

Method	PSO setting	
	Old	Constriction
LX-SSA-VNS	2.65	3.41
SSA-VNS	1.99	2.77
LX-SSA	5.65	6.06
SSA	4.81	5.34
PSO	5.04	1.53
DE	6.99	7.03
VNS	3.59	4.15
MS-SLSQP	5.28	5.71
PSO position	5	1
LX-SSA-VNS vs. PSO (W/T/L)	35/33/0	0/21/47
SSA-VNS vs. PSO (W/T/L)	41/27/0	0/22/46

Ranks: feasibility-aware rule (fewer than 15 of 30 feasible runs in a case = ranked last); bold: best average rank. W/T/L: cases in which the hybrid is significantly better / not different / worse than PSO (two-sided Wilcoxon signed-rank test, 30 seed-paired runs, Holm-adjusted over the seven comparisons of the hybrid in each case, $\alpha = 0.05$).

the final swarm is 8.9 times as wide and 13 times as fast, 8.7% of the coordinates are clipped (constriction: 1.6%), and only 2.6% of the particle evaluations in the second half of a run are feasible (48.4%). Its infeasible final layouts mostly violate only the spacing constraint (246 of 282): as g^b is in m^2 and g^s in m , the best particle lies inside the farm, with turbines too close. In Table II, the old setting would place PSO fifth (average rank 5.12), not second.

Table I shows the effect on a comparison: it ranks the methods of Table II, with LX-SSA-VNS in place of PSO-VNS, once with each PSO setting; all other methods use identical runs in both columns. With the old setting, SSA-VNS has the best average rank and PSO ranks behind a salp-swarm hybrid; with the constriction setting, PSO ranks ahead of every salp-swarm method, and the W/T/L of SSA-VNS against PSO changes from 41/27/0 to 0/22/46. The constriction coefficients are standard values [43], not tuned on these cases, nor are those of the other methods. The old setting was run only at 6,030 evaluations; with more evaluations the order changes again: on the six largest cases at 120,030 evaluations, SSA-VNS ranks ahead of PSO (average ranks 4.17 and 5.83; Section VIII-A). A PSO baseline that violates (8) thus makes methods compared with it, including our own LX-SSA, appear stronger than they are; hence PSO is the first phase of PSO-VNS.

B. Overall Comparison

Over the eight methods (Table II), the Friedman test rejects the hypothesis of equal performance ($\chi_F^2 = 319.3$, $p = 4.6 \times 10^{-65}$; Iman–Davenport $F_F = 136.5$, $p = 6.7 \times 10^{-109}$). PSO-VNS has the best average rank (1.82) and alone ranks first in 29 cases. In the Holm-adjusted post hoc tests (p_z in Table II; tied ranks make them conservative), PSO-VNS ranks significantly better than SSA-VNS, VNS, SSA, MS-SLSQP, LX-SSA and DE, but not than PSO ($p_{\text{Holm}} = 0.71$). $\Delta\bar{L}$

TABLE II

CASE-LEVEL ANALYSIS OVER THE 68 BENCHMARK CASES. AVG. RANK: AVERAGE RANK OF THE MEAN FEASIBLE OBJECTIVE (1 = BEST; METHODS WITH FEWER THAN 15 FEASIBLE RUNS IN A CASE RANKED LAST); “BEST”: CASES IN WHICH THE METHOD ALONE RANKS FIRST; “FEAS.”: FEASIBLE RUNS (%); p_z : HOLM-ADJUSTED p OF THE AVERAGE-RANK TEST AGAINST PSO-VNS. BECAUSE MEAN-RANK TESTS DEPEND ON THE POOL OF METHODS [39], p_W AND $\Delta\bar{L}$ GIVE THE TWO-SIDED WILCOXON SIGNED-RANK TEST ON THE 68 PER-CASE MEAN WAKE LOSSES (HOLM-ADJUSTED OVER THE 7 METHODS) AND THE MEAN WAKE-LOSS DIFFERENCE OVER THE CASES IN WHICH BOTH METHODS HAVE AT LEAST 15 FEASIBLE RUNS (66 FOR SSA, 66 FOR LX-SSA, 50 FOR DE, OTHERWISE 68; PERCENTAGE POINTS, PSO-VNS MINUS METHOD; NEGATIVE = PSO-VNS BETTER)

Method	Avg. rank	Best	Feas.	p_z	p_W	$\Delta\bar{L}$
PSO-VNS	1.82	29	100.0	—	—	—
PSO	1.97	25	100.0	0.713	0.296	−0.018
SSA-VNS	3.45	0	99.6	2.0×10^{-4}	5.7×10^{-11}	−0.330
VNS	4.40	3	100.0	2.4×10^{-9}	1.3×10^{-10}	−0.465
SSA	5.38	0	97.6	8.3×10^{-17}	4.5×10^{-11}	−0.611
MS-SLSQP	5.88	1	100.0	1.8×10^{-21}	4.2×10^{-11}	−0.435
LX-SSA	6.06	0	96.9	3.3×10^{-23}	4.5×10^{-11}	−0.765
DE	7.04	0	71.4	1.0×10^{-34}	5.6×10^{-11}	−0.887

Friedman $\chi_F^2 = 319.3$ (7 d.f.), $p = 4.6 \times 10^{-65}$;

Iman–Davenport $F_F = 136.5$ (7, 469 d.f.), $p = 6.7 \times 10^{-109}$

TABLE III

PAIRWISE OUTCOME OF PSO-VNS AGAINST EACH METHOD: CASES IN WHICH PSO-VNS IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (TWO-SIDED WILCOXON SIGNED-RANK TEST, 30 SEED-PAIRED RUNS, HOLM-ADJUSTED OVER THE SEVEN COMPARISONS OF PSO-VNS IN EACH CASE, A DIFFERENT FAMILY FROM TABLE IV, $\alpha = 0.05$; AN INFEASIBLE RUN RANKS BELOW EVERY FEASIBLE RUN); LAST ROW: MEDIAN RANK-BISERIAL CORRELATION (POSITIVE = PSO-VNS BETTER; ZERO DIFFERENCES DROPPED, $r_{\text{rb}} = 0$ IF ALL ARE ZERO)

DS	r (m)	Cases	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
I	500	9	0/8/1	6/3/0	8/1/0	8/1/0	7/2/0	6/3/0	7/2/0
I	750	11	1/8/2	9/2/0	10/1/0	10/1/0	9/2/0	8/3/0	10/1/0
I	1000	14	3/11/0	9/5/0	12/2/0	12/2/0	12/2/0	12/2/0	12/2/0
II	500	9	3/5/1	7/2/0	7/2/0	8/1/0	7/2/0	7/1/1	8/1/0
II	750	11	2/9/0	8/3/0	10/1/0	10/1/0	9/2/0	8/3/0	9/2/0
II	1000	14	5/9/0	10/4/0	12/2/0	12/2/0	12/2/0	10/4/0	13/1/0
All		68	14/50/4	49/19/0	59/9/0	60/8/0	56/12/0	51/16/1	59/9/0
$\bar{r}_{\text{rb}}$			+0.08	+0.72	+0.98	+0.97	+1.00	+0.84	+0.99

averages only over the cases in which both methods have at least 15 feasible runs (fewer than 68 for SSA, LX-SSA and DE). The rank order and the main conclusions are unchanged for thresholds of 1, 10, 15, 20, 25 and 30 feasible runs, significant comparisons with PSO-VNS favor the same method in each of the six clusters, and leaving out any one cluster changes none of them (PSO-VNS always first; Tables S15–S18).

The run-level W/T/L counts (Table III; Holm-adjusted within each case over the seven comparisons with PSO-VNS) are descriptive. Against PSO, PSO-VNS is significantly better in 14 cases, tied in 50 and worse in 4. On the case means the two are practically equivalent at ± 0.05 percentage points: $\Delta\bar{L} = -0.018$ (90% bootstrap CI $[-0.041, 0.004]$; TOST $p = 0.011$; minimal equivalence margin 0.041; Bayesian signed-rank $P(\text{ROPE}) > 0.99$; Table S13), 0.020% of the benchmark AEP, and PSO has the lower case mean in 26 of the 68 cases. Neither the per-case standard deviation nor the worst feasible

run differs significantly ($p = 0.55$ and 0.80). The advantage of PSO-VNS is concentrated in the larger layouts of Data Set II (post hoc split): with $N \geq 10$ it has the lower mean wake loss in 10 of 10 cases (Wilcoxon $p = 0.002$; $p = 0.018$ Holm-adjusted over all 28 case-mean tests of the paper; 0.221% of AEP), significantly more than with $N < 10$ (permutation within clusters, $p = 1.0 \times 10^{-4}$), and the gain grows with N within every Data Set II farm (mean Spearman $\rho = -0.66$, $p = 3.0 \times 10^{-4}$). Data Set I shows no such effect ($p = 0.29$; PSO is lower in 6 of the 10 cases with $N \geq 10$), and pooled over both data sets the $N \geq 10$ subgroup is not significant after the Holm correction ($p = 0.29$). For $N < 10$ both reach nearly the same wake loss (mean absolute difference 0.03 percentage points). Against SSA-VNS (a salp-swarm first phase), PSO-VNS is significantly better in 49 cases and never worse ($p_{\text{Holm}} = 5.7 \times 10^{-11}$). Against SSA, LX-SSA, DE, VNS and MS-SLSQP, it is significantly better in most cases; it is never significantly worse than SSA-VNS, SSA, LX-SSA, DE and MS-SLSQP and significantly worse than VNS in one case. PSO-VNS is thus a simple reference method with limits: it has the best average rank but is, on average, practically equivalent to PSO; it adds feasibility in every run (Horns Rev 1: 30/30 vs 19/30 for PSO, Section VIII-B) and gains in the larger Data Set II layouts. On the six largest cases it ranks best at all three budgets (exploratory; Section VIII-A). The order of the leading methods depends on the conditions: with feasible initial layouts, VNS ranks first (Section VIII-A), and with a Gaussian wake model, PSO ranks narrowly ahead of PSO-VNS (Section IX).

C. Solution Quality, Feasibility and Convergence

The wake loss grows with N , falls as the farm becomes larger, and is higher for the broader wind rose of Data Set II (Section S-II-D). On the six largest cases, the mean wake loss of PSO minus that of PSO-VNS is -0.18 to $+0.41$ percentage points (PSO-VNS lower in four). PSO-VNS ends feasible in all runs (others: Table II; DE 71.4%); the feasibility counts of PSO, SSA, LX-SSA and DE do not depend on the wind data set, as the penalty dominates F_p while every candidate is infeasible. In the densest cases (500 m, $N = 10$), the global best of the PSO phase is feasible at the switch in only 63.3% of the runs, but all runs are feasible after the VNS phase.

After the switch, PSO-VNS and PSO both keep improving (Fig. 1): over all feasible runs, the VNS phase removes on average 30.3% (median 22.6%) of the wake loss left at the end of the PSO phase and makes 26 runs infeasible at the switch feasible, whereas continuing PSO for the same 3,000 evaluations removes 29.2% (median 20.6%).

VII. COMPONENT ANALYSIS OF TWO-PHASE HYBRIDS

The variants of Table IV combine a Phase 1 (PSO, SSA, LX-SSA, random sampling or none) with or without the same VNS phase (6,030 evaluations, same seeds; convergence: Fig. 1); their average ranks differ (Friedman $p = 6.0 \times 10^{-76}$). Here p is from the Wilcoxon test on the 68 case means (unadjusted; Holm over the fifteen contrasts in Table IV) and $\Delta\bar{L}$ is their

TABLE IV
COMPONENT ANALYSIS OVER THE 68 BENCHMARK CASES (30 SEED-PAIRED RUNS, 6,030 EVALUATIONS). TOP: AVERAGE RANK AMONG THE NINE VARIANTS (FRIEDMAN $\chi^2_F = 374.3$, 8 D.F., $p = 5.96 \times 10^{-76}$) AND FEASIBLE RUNS. BOTTOM: PLANNED CONTRASTS; W/T/L: CASES IN WHICH THE FIRST VARIANT IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (RUN-LEVEL WILCOXON, HOLM-ADJUSTED OVER THE FIFTEEN CONTRASTS OF EACH CASE, A DIFFERENT FAMILY FROM TABLE III, $\alpha = 0.05$); p_W : WILCOXON TEST ON THE 68 PER-CASE MEAN WAKE LOSSES, HOLM-ADJUSTED OVER THE FIFTEEN CONTRASTS; $\Delta\bar{L}$: MEAN DIFFERENCE OF THE CASE-MEAN WAKE LOSSES (PERCENTAGE POINTS; NEGATIVE = FIRST VARIANT BETTER)

Variant	Phase 1	Phase 2	Avg. rank	Feas. (%)
PSO-VNS	PSO	VNS	1.96	100.0
SSA-VNS	SSA	VNS	4.46	99.6
LX-SSA-VNS	LX-SSA	VNS	5.41	99.9
RS-VNS	random (square)	VNS	5.29	99.4
RSD-VNS	random (disc)	VNS	3.77	99.8
VNS	–	VNS	6.12	100.0
PSO	PSO	–	2.24	100.0
SSA	SSA	–	7.50	97.6
LX-SSA	LX-SSA	–	8.25	96.9

Contrast	Isolates	W/T/L	p_W	$\Delta\bar{L}$
PSO-VNS vs. PSO	VNS phase (PSO)	8/58/2	0.592	−0.018
SSA-VNS vs. SSA	VNS phase (SSA)	39/29/0	5.7×10^{-10}	−0.294
LX-SSA-VNS vs. LX-SSA	VNS phase (LX-SSA)	34/34/0	1.1×10^{-10}	−0.394
PSO-VNS vs. VNS	swarm start vs. best initial point	49/18/1	5.7×10^{-10}	−0.465
PSO-VNS vs. RS-VNS	PSO vs. random sampling	47/21/0	3.2×10^{-10}	−0.399
SSA-VNS vs. RS-VNS	SSA vs. random sampling	2/66/0	0.002	−0.068
LX-SSA-VNS vs. RS-VNS	LX-SSA vs. random sampling	1/67/0	0.718	−0.005
PSO-VNS vs. RSD-VNS	PSO vs. disc sampling	45/23/0	4.0×10^{-10}	−0.306
SSA-VNS vs. RSD-VNS	SSA vs. disc sampling	0/66/2	0.014	+0.024
LX-SSA-VNS vs. RSD-VNS	LX-SSA vs. disc sampling	1/62/5	8.2×10^{-6}	+0.087
RSD-VNS vs. RS-VNS	disc vs. square sampling	6/62/0	4.2×10^{-7}	−0.093
PSO-VNS vs. SSA-VNS	PSO vs. SSA as Phase 1	44/24/0	2.5×10^{-10}	−0.330
PSO-VNS vs. LX-SSA-VNS	PSO vs. LX-SSA as Phase 1	47/21/0	2.3×10^{-10}	−0.393
SSA-VNS vs. LX-SSA-VNS	Laplace step in the hybrid	2/66/0	2.7×10^{-5}	−0.063
LX-SSA vs. SSA	Laplace step alone	0/64/4	1.3×10^{-6}	+0.153

mean difference in pp; it is practically equivalent to zero if its 90% bootstrap interval (in brackets) lies within ± 0.05 pp.

VNS phase. The VNS phase significantly improves both salp-swarm methods: SSA-VNS is 39/29/0 (W/T/L) against SSA and LX-SSA-VNS 34/34/0 against LX-SSA ($\Delta\bar{L} = -0.328$ and -0.431). The PSO contrast is clean, as PSO-VNS and PSO share their first 3,030 evaluations (Section IV): switching to VNS gives a small gain ($\Delta\bar{L} = -0.018$, $p = 0.30$) that is practically equivalent to zero ($[-0.041, 0.004]$). At this budget the VNS phase is essentially one compass-search descent: for PSO-VNS, the first descent yields 97.5% of the Phase-2 gain, shaking uses 0.02% of the evaluations, and the median run with $N \geq 12$ never shakes (Table S36). The analysis thus measures the value of a local-search phase; the VNS neighborhoods can matter only for small N .

Swarm phase. Three controls isolate Phase 1: VNS alone and random sampling of equal cost, uniform in the site

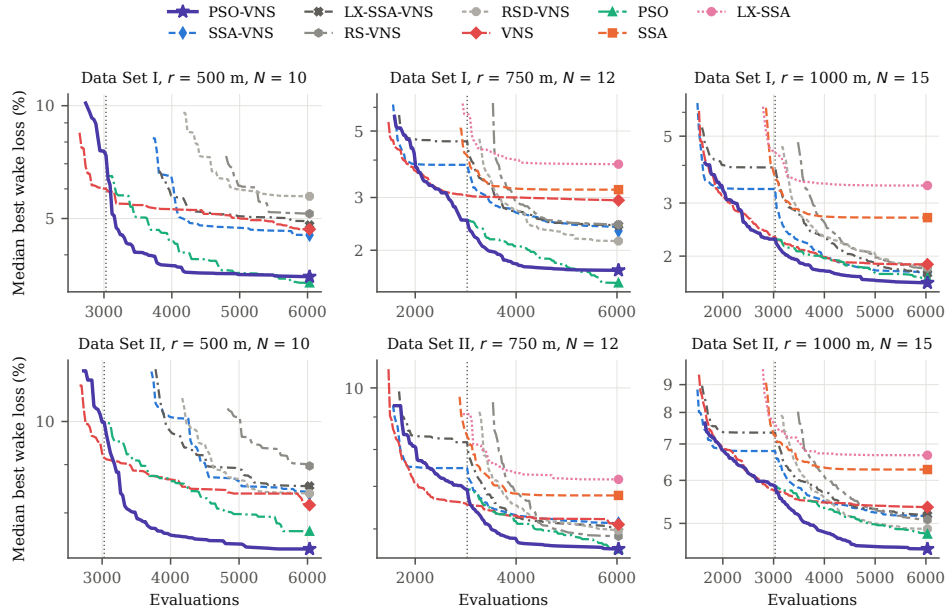


Fig. 1. Convergence of the nine component-analysis variants (Table IV) for the largest turbine count of each farm: median best feasible wake loss over 30 runs versus evaluations. PSO-VNS and PSO coincide up to 3,030 evaluations (dotted line), where PSO-VNS switches to VNS. Curves of the main comparison: Fig. S3 of the supplementary material.

disc (RSD-VNS) or in its bounding square (RS-VNS). RS-VNS is weak: a square sample has all N turbines inside the circle only with probability $(\pi/4)^N$, so 730 of its 2,040 runs draw no feasible sample (RSD-VNS: 494); RSD-VNS beats it (6/62/0, $\Delta\bar{L} = -0.093$) and ranks third (3.77). A PSO phase clearly beats every control: PSO-VNS is 45/23/0 against RSD-VNS ($\Delta\bar{L} = -0.306$), 47/21/0 against RS-VNS (-0.399) and 49/18/1 against VNS. Salp-swarm phases add nothing beyond disc sampling: against RSD-VNS, the SSA phase gives no gain (SSA-VNS: 0/66/2; the case means, not the clusters, favor RSD-VNS: $+0.024$), and LX-SSA-VNS is 1/62/5 and worse ($+0.087$, $p = 1.4 \times 10^{-6}$). The small SSA gain over RS-VNS (2/66/0, -0.068 , $p = 4.5 \times 10^{-4}$) thus only reflects that control's weakness; LX-SSA-VNS and RS-VNS are practically equivalent ($[-0.034, 0.021]$).

Laplace step. The Laplace step makes the salp swarm worse. Without VNS, LX-SSA is 0/64/4 against SSA ($\Delta\bar{L} = +0.166$, $p = 1.9 \times 10^{-7}$); in the hybrid, SSA-VNS is 2/66/0 against LX-SSA-VNS and better on the case means ($\Delta\bar{L} = -0.063$, $p = 5.4 \times 10^{-6}$). As LX-SSA costs $2N_p$ evaluations per iteration (Section III-B) and so runs half as many iterations as SSA, these contrasts do not separate the Laplace step from this cost.

Choice of Phase 1. With the same VNS phase, PSO-VNS is 44/24/0 against SSA-VNS and 47/21/0 against LX-SSA-VNS. At the switch, its global best is feasible more often (98.7% of the runs vs. 95.7% and 94.1%) and, when feasible, has a lower wake loss (1.63% vs. 2.01% and 2.11%), so VNS starts from a better layout. Whether PSO helps mainly by faster feasibility or by better exploration remains open.

Budget split. Table V compares the preset split $\omega = 0.5$, used for all other results, with $\omega = 0.25$, 0.75, 0.9 and $\omega = 1$ (plain PSO) on the twelve cases with the middle and the largest N of each farm (per case: Table S14). With $\omega = 0.75$, the

TABLE V
BUDGET SPLIT OF PSO-VNS: SHARE ω OF THE 6,030 EVALUATIONS FOR PSO ($\omega = 0.25, 0.5, 0.75, 0.9$; $\omega = 1$: PSO ALONE, SAME SEEDS) ON 12 CASES (MIDDLE AND LARGEST N OF EACH FARM AND DATA SET, 30 SEEDS): MEAN WAKE LOSS, AVERAGE RANK AMONG THE FIVE SETTINGS, AND CASES IN WHICH THE DEFAULT $\omega = 0.5$ IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (RUN-LEVEL WILCOXON, HOLM-ADJUSTED OVER THE SIX COMPARISONS OF EACH CASE)

ω	Mean loss (%)	Avg. rank	0.5 vs. ω (W/T/L)
0.25	2.796	4.75	5/7/0
0.5	2.643	3.25	—
0.75	2.543	1.42	0/8/4
0.9	2.553	1.83	0/8/4
1 (PSO alone)	2.693	3.75	3/9/0

$\omega = 0.75$ vs. $\omega = 1$ (PSO alone): 9/3/0 (W/T/L from the $\omega = 0.75$ side)
 $\omega = 0.9$ vs. $\omega = 0.75$: 0/12/0 (W/T/L from the $\omega = 0.9$ side)

mean wake loss is lower than with $\omega = 0.5$ in eleven cases (significantly in four; 0.10 pp on average), and significantly higher in no case; $\omega = 0.25$ is significantly worse than the default in five cases and better in no case. $\omega = 0.9$ does not beat $\omega = 0.75$ (W/T/L 0/12/0, $p = 0.15$). Plain PSO has a higher mean loss than $\omega = 0.75$ (2.693% vs. 2.543%, rank 3.75; W/T/L of $\omega = 0.75$: 9/3/0), so the best tested split is not the PSO end point. Chosen on twelve cases only, a split above 0.5 is a candidate to be confirmed on the target problem; we do not re-tune ω for the main comparison, which may slightly understate PSO-VNS.

VIII. TESTS BEYOND THE BENCHMARK

A. Budget Scaling and Feasibility-Preserving Initialization

Budget. On the six largest benchmark cases, PSO-VNS keeps the best average rank at all three budgets (6,030, 30,030

TABLE VI

SIX LARGEST BENCHMARK CASES: MEAN WAKE LOSS (%) AND FEASIBLE RUNS (%) AT 6,030 EVALUATIONS WITH RANDOM AND FEASIBILITY-PRESERVING INITIALIZATION, AND AVERAGE RANK AT 6,030, 30,030 AND 120,030 EVALUATIONS (RANDOM INITIALIZATION; 30 SEEDS); N/R: NOT RUN

Method	Random init.		Feasible init.		Avg. rank		
	Loss	Feas.	Loss	Feas.	6,030	30,030	120,030
PSO-VNS	4.084	100.0	3.938	100.0	1.50	1.17	2.17
SSA-VNS	4.678	95.0	3.955	100.0	4.33	4.17	4.17
LX-SSA-VNS	4.802	98.9	3.952	100.0	5.17	5.67	5.83
RS-VNS	4.830	93.3	n/r	n/r	4.67	6.33	4.67
VNS	4.950	100.0	3.810	100.0	6.50	4.83	3.17
PSO	4.181	100.0	5.549	100.0	1.83	2.50	5.83
SSA	4.828	80.0	4.947	100.0	8.33	8.67	8.83
LX-SSA	5.379	77.8	5.048	100.0	8.67	9.83	10.00
DE	—	1.1	5.549	100.0	10.00	5.17	3.33
MS-SLSQP	4.522	99.4	4.294	100.0	4.00	6.67	7.00

and 120,030 evaluations: 1.50, 1.17 and 2.17; Table VI; Fig. S9); it ranks first in 4, 5 and 3 of the six cases (exploratory; rank differences not tested). PSO's mean wake loss minus that of PSO-VNS is +0.10, +0.18 and +0.27 percentage points (per case: Table S25). The order of the other methods depends on the budget. PSO falls back from an average rank of 1.83 to 5.83, so that at 120,030 evaluations SSA-VNS ranks ahead of it (4.17), while VNS (6.50 to 3.17) and DE move up. At 120,030 evaluations, VNS, SSA-VNS, RS-VNS, LX-SSA-VNS and DE all come within 0.17 percentage points of the mean wake loss of PSO-VNS (3.14%).

Feasible starts. With feasibility-preserving initialization, the share of feasible runs rises for SSA-VNS (95.0% to 100.0%), SSA (80.0% to 100.0%), LX-SSA (77.8% to 100.0%) and DE (1.1% to 100.0%). VNS alone is then best in mean wake loss and average rank; PSO-VNS follows (rank 2.83, loss 3.94%; 4.08% with random starts; Table S24). PSO and DE never improve on the best initial layout (210 runs each).

Proposition 3: From feasible starts with $\mathbf{V}^i = \mathbf{0}$ and $\mathbf{P}^i = \mathbf{X}^i$, the particle holding $\mathbf{G}$ keeps $\mathbf{V} = \mathbf{0}$ until $\mathbf{G}$ changes, and a PSO or DE trial replaces $\mathbf{G}$ only if it is feasible and has a lower wake loss.

In our runs, no trial passes this test (proofs: Section S-VI-C; data: Table S37): apart from the re-evaluations of $\mathbf{G}$, all PSO candidates and all 126,000 DE trials are infeasible, and no personal best ever changes. PSO moves each coordinate by an independent random weight, with overshoot, toward the same coordinate of $\mathbf{G}$, and DE takes most coordinates from a combination of three other layouts; for distinct dense layouts, both make turbines collide (median minimum spacing at the first move 95 m, limit 308 m), and most trials also place a turbine outside the site. Relabelling the turbines to match those of $\mathbf{G}$ does not help (0.1% of first PSO moves feasible). The stall is observed, not implied; by the proposition, only a feasible, better trial can end it. VNS, which moves one coordinate at a time, is not affected. From random starts, PSO may thus help PSO-VNS mainly by reaching feasibility rather than by a better wake-loss search, and the advantage of PSO-VNS depends on the initialization.

TABLE VII

HORNS REV 1 16-TURBINE BLOCK (WAKE-FREE AEP 148.81 GWh/YR): MEAN AEP (GWh/YR) OVER THE FEASIBLE RUNS OF 30 SEEDS AT 6,030 EVALUATIONS, FEASIBLE RUNS, HOLM-ADJUSTED WILCOXON p OF PSO-VNS VS. EACH METHOD, AND MEAN AEP LOSS (%) AT 6,030 EVALUATIONS WITH RANDOM (R) AND FEASIBILITY-PRESERVING (F) INITIALIZATION AND AT 30,030 AND 120,030 EVALUATIONS (R; 10 SEEDS AT 120,030); SUPERScript: FEASIBLE RUNS WHEN NOT ALL RUNS ARE FEASIBLE

Layout / method	6,030 evaluations, R			Loss (%)			
	AEP	Feas.	p_{Holm}	6k R	6k F	30k	120k
Installed layout	139.82	—	—	6.04	—	—	—
PSO-VNS	138.29	30/30	—	7.07	6.97	6.52	6.40
SSA-VNS	137.59	30/30	2.6×10^{-4}	7.54	6.98	7.19	6.47
LX-SSA-VNS	137.41	30/30	1.6×10^{-4}	7.66	7.02	7.07	6.44
RS-VNS	137.58	28/30	1.1×10^{-4}	7.54 ²⁸	n/r	7.04	6.25
VNS	137.35	30/30	1.1×10^{-4}	7.70	6.91	6.85	6.23
PSO	137.79	19/30	4.6×10^{-5}	7.40 ¹⁹	8.35	6.78 ²⁸	6.63
SSA	136.25	30/30	2.1×10^{-7}	8.44	7.83	8.17	7.62
LX-SSA	136.00	28/30	6.5×10^{-8}	8.61 ²⁸	7.89	8.50	8.28
DE	—	0/30	1.7×10^{-8}	—	8.35	7.40 ⁸	6.12
MS-SLSQP	136.31	30/30	1.7×10^{-8}	8.40	8.00	7.82	7.31

B. Horns Rev 1 16-Turbine Block

The model (Section S-I-G) uses the Jensen wake with a hub-center in-wake test and 5° direction and 1-m/s speed bins; the boundary violation in (4) is the distance (m) outside the parallelogram. For the 80-turbine farm, our AEP agrees with PyWake [54] within 0.64%. The installed layout, a regular $7D$ grid, reaches 139.82 GWh/yr (wake loss 6.04%; Table VII; layouts: Fig. S10). At 6,030 evaluations, PSO-VNS is feasible in 30/30 runs, PSO in only 19/30 (30/30 with feasible starts); PSO-VNS has the highest mean AEP, 138.29 GWh/yr (SD 0.75); in the seed-paired run-level test (infeasible runs ranked last) it is significantly better than every other method ($p_{\text{Holm}} \leq 2.6 \times 10^{-4}$). Its best run reaches 139.78 GWh/yr, and 0 of its 30 runs exceed the installed layout. At 30,030 evaluations, PSO-VNS still has the highest mean AEP, 139.11 GWh/yr (loss 6.52%), and 3 of its 30 runs exceed the installed layout. At 120,030 evaluations (10 seeds), its loss of 6.40% is no longer the lowest; DE has the highest mean AEP. The optimized layouts may use the benchmark spacing $4D$, whereas the installed grid keeps $7D$ (no run with $7D$ was made); the block is evaluated without the wakes of the other 64 turbines; and cables, foundations, loads and navigation, which also shaped the installed grid, are not modeled. The comparison therefore tests the optimizers; it does not show that the installed layout could be improved.

C. IEA37 Case Study 1

In IEA37 Case Study 1 the participants chose their own optimizers and budgets [32]. None of our methods reaches the best feasible published layouts (Table VIII; layouts: Section S-IV-C). At 30,030 evaluations, the relative AEP difference to them is -1.33% (16 turbines) and -4.20% (36 turbines) for the best PSO-VNS layout and -3.90% and -6.51% for the mean over the 30 runs; the best PSO-VNS layout would rank third among 9 and seventh among 8 feasible submissions. The gap is larger for 36 turbines, where the best of our layouts comes from MS-SLSQP. As the participants' budgets are not

TABLE VIII
IEA37 CASE STUDY 1, 16- AND 36-TURBINE SCENARIOS: AEP IN MWH
OF THE BEST RUN OF EACH METHOD AT 6,030 AND 30,030
EVALUATIONS (30 SEEDS), COMPARED WITH THE EXAMPLE LAYOUT
AND THE BEST FEASIBLE PUBLISHED LAYOUT [32], [55]

Method / source	16 turbines ($r = 1300$ m)		36 turbines ($r = 2000$ m)	
	6,030	30,030	6,030	30,030
Example layout				
Best feasible published ^a	366,941.6		737,883.1	
	418,924.4		863,676.3	
PSO-VNS	402,782.4	413,331.9	787,708.9	827,385.0
PSO	403,753.1	412,702.4	787,988.7	830,256.4
SSA-VNS	403,251.3	406,924.6	766,803.3	819,930.5
SSA	395,150.2	401,459.3	750,707.4	802,538.2
LX-SSA	393,284.6	398,770.0	746,301.3	789,311.3
DE	357,391.3	397,402.7	—	—
VNS	400,169.3	404,076.0	785,791.3	824,230.2
MS-SLSQP	405,433.8	410,349.5	748,571.1	833,365.0

^aParticipant 4 in both scenarios (SNOPT with wake expansion continuation [TBD: verify against Baker et al. (2019)]). A higher submitted AEP places turbines up to 3.5 m outside the boundary and is not counted.

TABLE IX
ROBUSTNESS TO THE BENCHMARK MODEL: ALL FEASIBLE FINAL
LAYOUTS RE-EVALUATED (NOT RE-OPTIMIZED). Δ : MEAN RELATIVE
CHANGE OF THE OBJECTIVE (%); RANK: AVERAGE RANK OF PSO-VNS
OVER THE 68 CASES; BEST: BEST-RANKED METHOD; $\bar{\tau}$: MEAN
KENDALL CORRELATION BETWEEN THE BENCHMARK AND
ALTERNATIVE ORDERINGS WITHIN A CASE; SAME: CASES (%) WITH
UNCHANGED BEST METHOD. THE RANKS OF ALL METHODS ARE GIVEN
IN TABLE S30

Model	Δ (%)			Best	$\bar{\tau}$	Same
	DS I	DS II	Rank			
Benchmark	+0.0	+0.0	1.82	PSO-VNS	—	—
Cubic curve	-17.7	-36.9	1.82	PSO-VNS	0.95	94
Cubic, 25 m/s cut-out	-21.5	-37.0	1.80	PSO-VNS	0.95	93
Gaussian wake	-0.3	-0.4	1.97	PSO	0.66	54

reported comparably, the size of the gap is indicative only. The 64-turbine scenario was not run, and, like the published layouts, ours are optimized for 16 discrete directions and one wind speed.

D. Computational Cost and Published Results

Per evaluation, the metaheuristics need 0.23–0.45 ms and MS-SLSQP 0.53 ms, about 2.3 times as long as PSO (indicative; Table S29); one 6,030-evaluation run of PSO-VNS takes 1.5–3.2 s, depending on N . Published Kusiak–Song results [5]–[8] are discussed in Section S-IX; they are not compared numerically because budgets, constraint handling and reporting differ.

IX. SENSITIVITY TO THE MODELING ASSUMPTIONS

Power curve and wake model. To test whether our conclusions depend on the benchmark models, we re-evaluated every feasible final layout of the main comparison, without re-optimization, with 1) a cubic power curve [56], $f_c(s) = P_{\text{rated}}(s^3 - s_{\text{cut-in}}^3)/(s_{\text{rated}}^3 - s_{\text{cut-in}}^3)$ between the cut-in and rated speeds, 2) the same curve with a 25 m/s cut-out, and 3) the Gaussian wake model of [57] with growth rate $K_G = 0.04$ and the superposition and Weibull scaling of the benchmark (Section S-V-A of the supplementary material). The absolute

energy depends strongly on the model (Table IX): the cubic curve gives 17.7% (Data Set I) and 36.9% (Data Set II) less expected power than the linear benchmark curve, so our energy values are benchmark values, not AEP predictions for a specific turbine. The ranking of the methods is stable under the cubic curves ($\bar{\tau} = 0.95$; same best method in 94% of the cases), and PSO-VNS keeps the best average rank. It is less stable under the Gaussian model ($\bar{\tau} = 0.66$; same best method in 54% of the cases): this model differs from the Jensen cone both in its smooth lateral profile and in its (uncalibrated) expansion rate, so layouts tuned to the Jensen cone are judged differently. PSO ranks first and PSO-VNS second, but the margin in average rank is small and untested (1.91 vs. 1.97), both ahead of all other methods, as expected for two practically equivalent methods (Section X).

Spacing and boundary handling. The optimized layouts often lie on the 4D constraint inherited from the benchmark: of the feasible final layouts of the eight methods, 49% also satisfy 5D and 36% satisfy 6D (4.1% and 0.1% for $N = 11$ –15), so they cannot be transferred to sites with a larger spacing without re-optimization. At 5D and 6D, a multi-start packing search fits fewer turbines into each circle, and the largest 500-m cases may be infeasible (bounds and a re-optimization of six cases: Tables S31 and S32). The boundary rule also matters: an otherwise identical SSA that projects every turbine radially onto the circle instead of clipping the coordinates attains higher mean benchmark objectives than the original SSA in all six representative cases at 3,030 evaluations (Mann–Whitney test, $p \leq 0.0036$ in every case). Because the boundary rule alone changes the results, all methods use the same rule (Section II-C).

X. DISCUSSION AND RECOMMENDATIONS

In the WFLOP, which method appears best can depend on the protocol as strongly as on the methods. *Baseline configuration:* replacing the old setting, which violates the order-2 stability condition (8), by the constriction setting moves PSO from position 5 to 1 in Table I, ahead of every salp-swarm method at 6,030 evaluations. Instrumented runs show why: the old swarm stays 8.9 times wider, and only 2.6% of its particle evaluations are feasible (constriction: 48.4%). An advantage over PSO is informative only if the PSO baseline is convergent.

Component analysis: a swarm phase must also beat random sampling of equal cost, and the control must be strong. Sampling in the bounding square (RS-VNS) is weak (730 of its runs never sample a feasible layout; in the disc: 494). Against sampling in the disc (RSD-VNS), the SSA phase gives no gain (+0.024 pp) and the LX-SSA phase is worse (+0.087 pp), so the small SSA gain over RS-VNS only reflects that control's weakness; the PSO phase beats both clearly (−0.306 pp). At 6,030 evaluations, the second phase is essentially one compass-search descent (Section VII), so this analysis measures the value of a start for a local search, not of the VNS neighborhoods.

PSO and PSO-VNS: a well-configured PSO and PSO-VNS are practically equivalent on the case means (90% interval

$[-0.041, 0.004]$ pp; mean gain 0.020% of benchmark AEP), with no significant difference in spread; the rank flip under the Gaussian model is consistent with this. PSO-VNS adds feasibility reliability (Horns Rev 1: 30/30 vs. 19/30 feasible runs) and a gain in the larger Data Set II layouts ($N \geq 10$: lower loss in 10 of 10 cases, Holm-adjusted over all tests $p = 0.018$).

For benchmark-type problems with random starts, a constriction PSO is a sound first choice and PSO-VNS a more reliable one; from feasible starts, VNS alone is competitive. For WFLOP comparisons, we recommend [23], [24]:

- 1) use baseline settings with documented convergence or standard defaults (for PSO, (8) or (7)), report them, and verify their behavior with instrumented runs;
- 2) give all methods the same budget, counting the initial population and gradient calls, and pair runs by seed;
- 3) use feasibility-aware ranking and report feasible runs;
- 4) test case means and runs with a multiplicity correction [22], and report threshold and, for nested cases, cluster-level robustness;
- 5) test practical equivalence with a stated margin [25], [26]; non-significance is not equivalence;
- 6) in ablations of hybrids, include the local method alone and an in-site random-sampling control of equal cost;
- 7) report feasible-start and model-robustness checks; publish code and data.

XI. LIMITATIONS

The benchmark uses the top-hat Jensen wake, the linear Kusiak–Song power curve, coarse direction bins that optimized layouts can exploit, circular farms, a $4D$ spacing and a prescribed N . The 68 cases form six clusters with nested N , so the p-values describe this benchmark. The $N \geq 10$ split is post hoc, and the equivalence margin was set after the primary analysis, so we report the smallest margin at which equivalence holds (0.041 pp for PSO-VNS vs. PSO). At 6,030 evaluations, the VNS neighborhoods are hardly exercised; the budget study covers only six cases and Horns Rev 1, and the old PSO setting was run only at 6,030 evaluations. DE, VNS and $\omega = 0.5$ are untuned defaults. The pool lacks CMA-ES [58] or L-SHADE, and MS-SLSQP uses finite-difference gradients on a discontinuous objective. A feasible random search [59] could be a still stronger control than sampling in the disc. From feasible starts, VNS ranks first and PSO and DE stall (Section VIII-A; mechanism observed, not derived); our statements on PSO-VNS refer to random starts. The Horns Rev 1 comparison tests the optimizers, not the installed design (Section VIII-B); on IEA37, the best PSO-VNS layouts at 30,030 evaluations reach 98.7% (16 turbines) and 95.8% (36 turbines) of the AEP of the best feasible published layouts. Turbulence and fatigue loads (critical at $4D$), wake steering, blockage, terrain, cables, electrical losses and cost objectives are outside our scope; all AEP values are gross.

XII. CONCLUSION

Three findings emerge. First, the baseline configuration changes the conclusions: the old PSO setting, which violates

the order-2 stability condition and rarely samples feasible layouts, puts PSO behind salp-swarm hybrids, whereas the constriction setting puts it ahead of every salp-swarm method. Second, a PSO phase gives the second phase (here one compass-search descent) a clearly better start than random sampling, whereas salp-swarm phases add nothing beyond random sampling. Third, a well-configured PSO and PSO-VNS are practically equivalent on average (± 0.05 pp); PSO-VNS adds feasibility reliability and gains in the larger Data Set II layouts. It is thus a consistent reference method, not a superior optimizer. Future work will address larger budgets and farms, stronger controls, better wake models and economic objectives.

ACKNOWLEDGMENT

[TBD: acknowledgment, or delete this section]

DATA AVAILABILITY

The source code of all methods and of the benchmark evaluator, the per-run results of all studies (including the runs with the old PSO setting) and the analysis scripts are openly available at [TBD: Zenodo DOI]. Each run record gives the method, case, seed, final objective (AEP for Horns Rev 1), feasibility, final turbine coordinates and convergence curve, and the budget and initialization where these vary. The file `analysis/README_reproduce.md` lists the script that produces each table, figure and number of this paper and of its supplementary material, and `analysis/requirements.txt` pins the software versions. The Horns Rev 1 and IEA37 input data are taken from [54], [55]. The supplementary material gives the model details, the pseudocode of SSA and LX-SSA, the wind data, the per-case results, the coordinates of the best layouts and further figures.

REFERENCES

- [1] R. Srikakulapu and U. Vinatha, “Optimized design of collector topology for offshore wind farm based on ant colony optimization with multiple travelling salesman problem,” *J. Mod. Power Syst. Clean Energy*, vol. 6, no. 6, pp. 1181–1192, Nov. 2018, doi: 10.1007/s40565-018-0386-4.
- [2] P. Hou, J. Zhu, K. Ma, G. Yang, W. Hu, and Z. Chen, “A review of offshore wind farm layout optimization and electrical system design methods,” *J. Mod. Power Syst. Clean Energy*, vol. 7, no. 5, pp. 975–986, Sep. 2019, doi: 10.1007/s40565-019-0550-5.
- [3] G. Mosetti, C. Poloni, and B. Diviacco, “Optimization of wind turbine positioning in large windfarms by means of a genetic algorithm,” *J. Wind Eng. Ind. Aerodyn.*, vol. 51, no. 1, pp. 105–116, Jan. 1994, doi: 10.1016/0167-6105(94)90080-9.
- [4] S. A. Grady, M. Y. Hussaini, and M. M. Abdullah, “Placement of wind turbines using genetic algorithms,” *Renew. Energy*, vol. 30, no. 2, pp. 259–270, Feb. 2005, doi: 10.1016/j.renene.2004.05.007.
- [5] A. Kusiak and Z. Song, “Design of wind farm layout for maximum wind energy capture,” *Renew. Energy*, vol. 35, no. 3, pp. 685–694, Mar. 2010, doi: 10.1016/j.renene.2009.08.019.
- [6] Y. Eroğlu and S. U. Seçkiner, “Design of wind farm layout using ant colony algorithm,” *Renew. Energy*, vol. 44, pp. 53–62, Aug. 2012, doi: 10.1016/j.renene.2011.12.013.
- [7] Y. Eroğlu and S. U. Seçkiner, “Wind farm layout optimization using particle filtering approach,” *Renew. Energy*, vol. 58, pp. 95–107, Oct. 2013, doi: 10.1016/j.renene.2013.02.019.
- [8] J. C. Bansal and P. Farswan, “Wind farm layout using biogeography based optimization,” *Renew. Energy*, vol. 107, pp. 386–402, Jul. 2017, doi: 10.1016/j.renene.2017.01.064.

- [9] C. N. Elkinton, J. F. Manwell, and J. G. McGowan, "Algorithms for offshore wind farm layout optimization," *Wind Eng.*, vol. 32, no. 1, pp. 67–84, Jan. 2008, doi: 10.1260/030952408784305877.
- [10] P. Asaah, L. Hao, and J. Ji, "Optimal placement of wind turbines in wind farm layout using particle swarm optimization," *J. Mod. Power Syst. Clean Energy*, vol. 9, no. 2, pp. 367–375, Mar. 2021, doi: 10.35833/MPCE.2019.000087.
- [11] X. Ju and F. Liu, "Wind farm layout optimization using self-informed genetic algorithm with information guided exploitation," *Appl. Energy*, vol. 248, pp. 429–445, Aug. 2019, doi: 10.1016/j.apenergy.2019.04.084.
- [12] F. Bai, X. Ju, S. Wang, W. Zhou, and F. Liu, "Wind farm layout optimization using adaptive evolutionary algorithm with Monte Carlo tree search reinforcement learning," *Energy Convers. Manage.*, vol. 252, Art. no. 115047, Jan. 2022, doi: 10.1016/j.enconman.2021.115047.
- [13] S. V. Nagpal, M. V. Liu, and C. L. Anderson, "A comparison of deterministic refinement techniques for wind farm layout optimization," *Renew. Energy*, vol. 168, pp. 581–592, May 2021, doi: 10.1016/j.renene.2020.12.043.
- [14] D. Cazzaro and D. Pisinger, "Variable neighborhood search for large offshore wind farm layout optimization," *Comput. Oper. Res.*, vol. 138, Art. no. 105588, Feb. 2022, doi: 10.1016/j.cor.2021.105588.
- [15] E. Quaegebeur, R. Bos, and M. B. Zaaijer, "Wind farm layout optimization using pseudo-gradients," *Wind Energy Sci.*, vol. 6, no. 3, pp. 815–839, Jun. 2021, doi: 10.5194/wes-6-815-2021.
- [16] S. Tao, Q. Xu, A. Feijóo, G. Zheng, and J. Zhou, "Wind farm layout optimization with a three-dimensional Gaussian wake model," *Renew. Energy*, vol. 159, pp. 553–569, Oct. 2020, doi: 10.1016/j.renene.2020.06.003.
- [17] K. Yang, X. Deng, Z. Ti, S. Yang, S. Huang, and Y. Wang, "A data-driven layout optimization framework of large-scale wind farms based on machine learning," *Renew. Energy*, vol. 218, Art. no. 119240, Dec. 2023, doi: 10.1016/j.renene.2023.119240.
- [18] Z. Wang, Y. Tu, K. Zhang, Z. Han, Y. Cao, and D. Zhou, "An optimization framework for wind farm layout design using CFD-based Kriging model," *Ocean Eng.*, vol. 293, Art. no. 116644, Feb. 2024, doi: 10.1016/j.oceaneng.2023.116644.
- [19] P. Li, Y. Che, A. Hua, L. Wang, M. Zheng, and X. Guo, "A data-physics hybrid-driven layout optimization framework for large-scale wind farms," *Appl. Energy*, vol. 392, Art. no. 125908, Aug. 2025, doi: 10.1016/j.apenergy.2025.125908.
- [20] J. N. Hooker, "Testing heuristics: We have it all wrong," *J. Heuristics*, vol. 1, no. 1, pp. 33–42, 1995, doi: 10.1007/BF02430364.
- [21] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *J. Mach. Learn. Res.*, vol. 7, pp. 1–30, Jan. 2006.
- [22] J. Derrac, S. García, D. Molina, and F. Herrera, "A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms," *Swarm Evol. Comput.*, vol. 1, no. 1, pp. 3–18, 2011, doi: 10.1016/j.swevo.2011.02.002.
- [23] T. Bartz-Beielstein *et al.*, "Benchmarking in optimization: Best practice and open issues," 2020, *arXiv:2007.03488*.
- [24] A. LaTorre, D. Molina, E. Osaba, J. Poyatos, J. Del Ser, and F. Herrera, "A prescription of methodological guidelines for comparing bio-inspired optimization algorithms," *Swarm Evol. Comput.*, vol. 67, Art. no. 100973, Dec. 2021, doi: 10.1016/j.swevo.2021.100973.
- [25] A. Benavoli, G. Corani, J. Demšar, and M. Zaffalon, "Time for a change: A tutorial for comparing multiple classifiers through Bayesian analysis," *J. Mach. Learn. Res.*, vol. 18, no. 77, pp. 1–36, 2017.
- [26] D. Lakens, "Equivalence tests: A practical primer for t tests, correlations, and meta-analyses," *Soc. Psychol. Personal. Sci.*, vol. 8, no. 4, pp. 355–362, 2017, doi: 10.1177/1948550617697177.
- [27] S. Mirjalili, A. H. Gandomi, S. Z. Mirjalili, S. Saremi, H. Faris, and S. M. Mirjalili, "Salp swarm algorithm: A bio-inspired optimizer for engineering design problems," *Adv. Eng. Softw.*, vol. 114, pp. 163–191, Dec. 2017, doi: 10.1016/j.advengsoft.2017.07.002.
- [28] K. Sörensen, "Metaheuristics—The metaphor exposed," *Int. Trans. Oper. Res.*, vol. 22, no. 1, pp. 3–18, Jan. 2015, doi: 10.1111/itor.12001.
- [29] C. L. Camacho-Villalón, M. Dorigo, and T. Stützle, "Exposing the grey wolf, moth-flame, whale, firefly, bat, and antlion algorithms: Six misleading optimization techniques inspired by bestial metaphors," *Int. Trans. Oper. Res.*, vol. 30, no. 6, pp. 2945–2971, 2023, doi: 10.1111/itor.13176.
- [30] M. Castelli, L. Manzoni, L. Mariot, M. S. Nobile, and A. Tangherloni, "Salp swarm optimization: A critical review," *Expert Syst. Appl.*, vol. 189, Art. no. 116029, 2022, doi: 10.1016/j.eswa.2021.116029.
- [31] D. Wilson *et al.*, "Evolutionary computation for wind farm layout optimization," *Renew. Energy*, vol. 126, pp. 681–691, 2018, doi: 10.1016/j.renene.2018.03.052.
- [32] N. F. Baker, A. P. J. Stanley, J. J. Thomas, A. Ning, and K. Dykes, "Best practices for wake model and optimization algorithm selection in wind farm layout optimization," in *Proc. AIAA SciTech 2019 Forum*, San Diego, CA, USA, Jan. 2019, Paper AIAA 2019-0540, doi: 10.2514/6.2019-0540.
- [33] J. J. Thomas *et al.*, "A comparison of eight optimization methods applied to a wind farm layout optimization problem," *Wind Energy Sci.*, vol. 8, pp. 865–891, 2023, doi: 10.5194/wes-8-865-2023.
- [34] F. Azlan, J. C. Kurnia, B. T. Tan, and M.-Z. Ismadi, "Review on optimisation methods of wind farm array under three classical wind condition problems," *Renew. Sustain. Energy Rev.*, vol. 135, Art. no. 110047, Jan. 2021, doi: 10.1016/j.rser.2020.110047.
- [35] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. IEEE Int. Conf. Neural Netw. (ICNN)*, Perth, WA, Australia, Nov. 1995, vol. 4, pp. 1942–1948, doi: 10.1109/ICNN.1995.488968.
- [36] R. Poli, "Mean and variance of the sampling distribution of particle swarm optimizers during stagnation," *IEEE Trans. Evol. Comput.*, vol. 13, no. 4, pp. 712–721, Aug. 2009, doi: 10.1109/TEVC.2008.2011744.
- [37] M. R. Bonyadi and Z. Michalewicz, "Particle swarm optimization for single objective continuous space problems: A review," *Evol. Comput.*, vol. 25, no. 1, pp. 1–54, 2017, doi: 10.1162/EVCO_r_00180.
- [38] C. W. Cleghorn and A. P. Engelbrecht, "Particle swarm stability: A theoretical extension using the non-stagnate distribution assumption," *Swarm Intell.*, vol. 12, no. 1, pp. 1–22, 2018, doi: 10.1007/s11721-017-0141-x.
- [39] A. Benavoli, G. Corani, and F. Mangili, "Should we really use post-hoc tests based on mean-ranks?" *J. Mach. Learn. Res.*, vol. 17, no. 5, pp. 1–10, 2016.
- [40] N. Mladenović and P. Hansen, "Variable neighborhood search," *Comput. Oper. Res.*, vol. 24, no. 11, pp. 1097–1100, Nov. 1997, doi: 10.1016/S0305-0548(97)00031-2.
- [41] P. Hansen and N. Mladenović, "Variable neighborhood search: Principles and applications," *Eur. J. Oper. Res.*, vol. 130, no. 3, pp. 449–467, May 2001, doi: 10.1016/S0377-2217(00)00100-4.
- [42] P. Solanki and K. Deep, "Laplacian salp swarm algorithm for continuous optimization," *Int. J. Syst. Assur. Eng. Manage.*, [TBD: vol., no., pp.], 2023, doi: 10.1007/s13198-023-01935-y.
- [43] M. Clerc and J. Kennedy, "The particle swarm—Explosion, stability, and convergence in a multidimensional complex space," *IEEE Trans. Evol. Comput.*, vol. 6, no. 1, pp. 58–73, Feb. 2002, doi: 10.1109/4235.985692.
- [44] N. O. Jensen, "A note on wind generator interaction," Risø Nat. Lab., Roskilde, Denmark, Tech. Rep. Risø-M-2411, 1983.
- [45] I. Katić, J. Højstrup, and N. O. Jensen, "A simple model for cluster efficiency," in *Proc. Eur. Wind Energy Assoc. Conf. Exhib. (EWEC)*, Rome, Italy, Oct. 1986, vol. 1, pp. 407–410.
- [46] Y. Shi and R. Eberhart, "A modified particle swarm optimizer," in *Proc. IEEE Int. Conf. Evol. Comput. (ICEC)*, Anchorage, AK, USA, May 1998, pp. 69–73, doi: 10.1109/ICEC.1998.699146.
- [47] D. Bratton and J. Kennedy, "Defining a standard for particle swarm optimization," in *Proc. IEEE Swarm Intell. Symp. (SIS)*, Honolulu, HI, USA, Apr. 2007, pp. 120–127, doi: 10.1109/SIS.2007.368035.
- [48] D. Zaharie, "Critical values for the control parameters of differential evolution algorithms," in *Proc. MENDEL 2002, 8th Int. Conf. Soft Comput.*, Brno, Czech Republic, 2002, pp. 62–67.
- [49] K. Deep and M. Thakur, "A new crossover operator for real coded genetic algorithms," *Appl. Math. Comput.*, vol. 188, no. 1, pp. 895–911, May 2007, doi: 10.1016/j.amc.2006.10.047.
- [50] N. Mladenović, M. Dražić, V. Kovačević-Vujčić, and M. Čangalović, "General variable neighborhood search for the continuous optimization," *Eur. J. Oper. Res.*, vol. 191, no. 3, pp. 753–770, Dec. 2008, doi: 10.1016/j.ejor.2006.12.064.
- [51] R. Storn and K. Price, "Differential evolution—A simple and efficient heuristic for global optimization over continuous spaces," *J. Global Optim.*, vol. 11, no. 4, pp. 341–359, Dec. 1997, doi: 10.1023/A:1008202821328.
- [52] D. Kraft, "A software package for sequential quadratic programming," DFVLR, Inst. für Dynamik der Flugsysteme, Oberpfaffenhofen, Germany, Tech. Rep. DFVLR-FB 88-28, 1988.
- [53] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Stat.*, vol. 6, no. 2, pp. 65–70, 1979.
- [54] DTU Wind Energy, "PyWake: Open-source static wake modeling framework," Python package `py_wake`, ver. 2.6.20 (Horns Rev 1 site data in `py_wake/examples/data/hornsrev1.py`), MIT License, 2026. Accessed: Sep. 25, 2026. [Online]. Available: <https://gitlab.windenergy.dtu.dk/TOPFARM/PyWake>

- [55] N. F. Baker *et al.*, “IEA Wind Task 37 wind farm layout optimization case studies: Case files and submitted results,” GitHub repository, commit 267f6e5, 2020. Accessed: Sep. 28, 2026. [Online]. Available: <https://github.com/IEAWindSystems/iea37-wflo-casestudies>
- [56] C. Carrillo, A. F. Obando Montaña, J. Cidrás, and E. Díaz-Dorado, “Review of power curve modelling for wind turbines,” *Renew. Sustain. Energy Rev.*, vol. 21, pp. 572–581, May 2013, doi: 10.1016/j.rser.2013.01.012.
- [57] M. Bastankhah and F. Porté-Agel, “A new analytical model for wind-turbine wakes,” *Renew. Energy*, vol. 70, pp. 116–123, Oct. 2014, doi: 10.1016/j.renene.2014.01.029.
- [58] N. Hansen and A. Ostermeier, “Completely derandomized self-adaptation in evolution strategies,” *Evol. Comput.*, vol. 9, no. 2, pp. 159–195, 2001, doi: 10.1162/106365601750190398.
- [59] J. Feng and W. Z. Shen, “Solving the wind farm layout optimization problem using random search algorithm,” *Renew. Energy*, vol. 78, pp. 182–192, 2015, doi: 10.1016/j.renene.2015.01.005.

Prince Solanki [TBD: biography: degrees, institutions, research interests]

Pulkit Dwivedi [TBD: biography]

Vanita Garg [TBD: biography]

Vinay Shukla [TBD: biography]